$$\begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix},$$

where α is an unknown parameter. Plug this vector into the eigenvalue equation and solve for α in terms of θ. Why did we use a single parameter α? Notice that our suggested column vector must have unit length.

Here are the results:

$$\lambda_1 = 1$$

$$|\lambda_1\rangle = \begin{pmatrix} \cos\frac{\theta}{2} \\ \\ \sin\frac{\theta}{2} \end{pmatrix}$$

and

$$\lambda_2 = -1$$

$$|\lambda_2\rangle = \begin{pmatrix} -\sin\frac{\theta}{2} \\ \\ \cos\frac{\theta}{2} \end{pmatrix}.$$

Notice some important facts. First, the two eigenvalues are again +1 and −1. This should come as no surprise; the apparatus $\mathcal{A}$ can only give one of these two answers no matter which way it points. But it's good to see this come out of the equations. The second fact is that the two eigenvectors are orthogonal.

We are now ready to make an experimental prediction. Suppose $\mathcal{A}$ initially points along the *z* axis and that we prepare a spin in the *up* state |*u*>.